

TrES-3b

R-0580

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_240629 · created 2026-07-24T17:52:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460490.8738 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.056
Transit depth (Rp/Rs)^2	1.98 +/- 1.58 [%]
Orbital Inclination (inc)	79.73 +/- 2.86
Airmass coefficient 1 (a1)	4.86 +/- 2.34
Airmass coefficient 2 (a2)	-1.22 +/- 0.9
Scatter in the residuals of the lightcurve fit is	44.88 %
Best Comparison Star	None
Optimal Aperture	2.17
Optimal Annulus	4.5
Transit Duration (day)	0.025 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.141 $\pm$ 0.056 vs 0.16309 (deviation 0.37 σ)
Duration fit vs published	0.025 d vs 0.0567 d (deviation 1.1 σ)
Mid-time O–C	1.3 min
Depth SNR	2.4
Residual scatter	44.88 %

Light curve

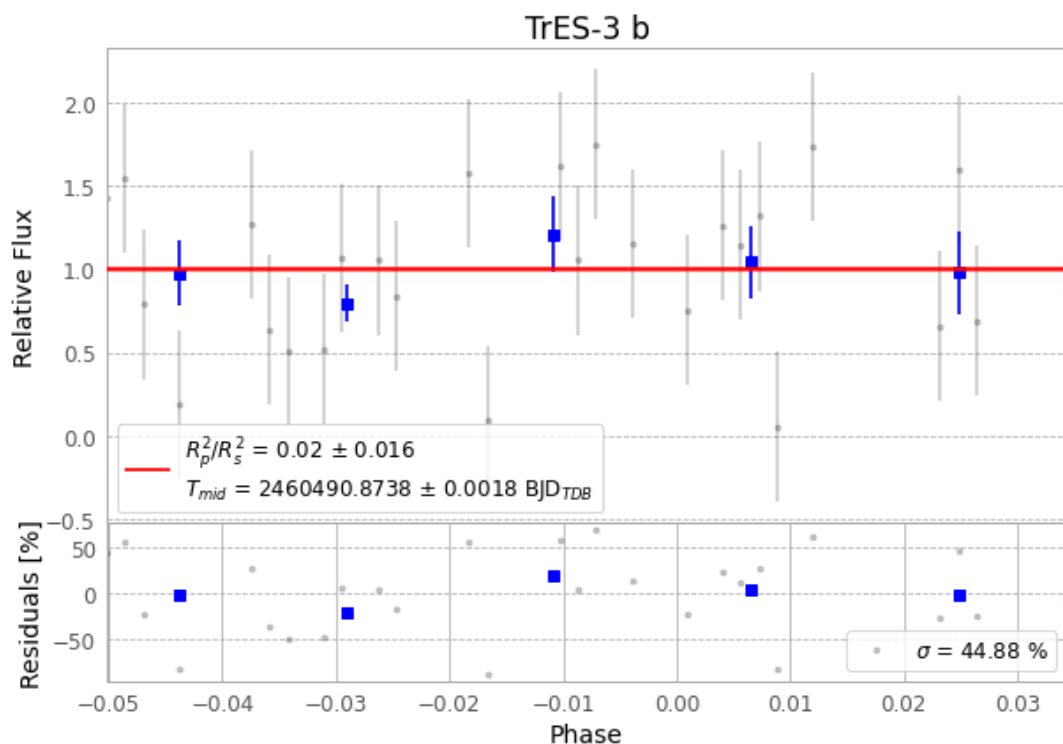


Figure 1 — FinalLightCurve_TrES-3 b_2024-06-29.png (SHA-256 5ddeb67e616ca13a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [545, 265], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1b277a47a8c42d17...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

56 FITS frames (37 MB), TRES-3240629071518.FITS → TRES-3240629100016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-240629.log (4d9e5edf7d67...), FinalLightCurve_TrES-3 b_2024-06-29.png (5ddeb67e616c...), FinalLightCurve_TrES-3 b_2024-06-29.csv (8d6160d092d0...)

Compute resources

Wall time 100.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 17:52Z.